

ABHISHEK AGRAHARI

Embedded Software Engineer · EV Firmware, Telematics & Vehicle Intelligence

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PROFESSIONAL SUMMARY

Embedded Software Engineer with 1.5+ years shipping production firmware on TI MCU, STM32 and ESP32. Currently building automotive-grade Vehicle Intelligence Modules at Vecmocon — BMS, motor controller and VCU over CAN, Quectel EC200 cellular uplink. Previously owned products end-to-end from schematic to commercial deployment: 2,000+ units validated, field issues cut by 90%. Strong in FreeRTOS design, bootloaders/FOTA, low-power design and hardware bring-up; building expertise in AIS-140.

TECHNICAL SKILLS

EV & Automotive	CAN / CAN-FD, BMS & VCU integration, motor-controller comms, vehicle telematics & GPS tracking, AIS-140, FOTA / OTA
Microcontrollers	TI MCU, STM32 (F1 / F4), ESP32 / ESP32-S3, ARM Cortex-M, ATmega328P, 8051
Languages / RTOS	C, C++, Python, Assembly · FreeRTOS, CMSIS-RTOS, embOS, bare-metal
Connectivity	UART, SPI, I2C, CAN, BLE, MQTT, HTTP, NB-IoT · Quectel EC200-series, Neoway N58 cellular modules
Firmware	Bootloaders, interrupt-driven design, DMA, multi-threading, memory optimisation, low-power, peripheral drivers
Debug & Tools	JTAG / SWD, logic analyser, oscilloscope, CAN analyser, STM32CubeIDE, Keil µVision, Git, Jira

WORK EXPERIENCE

Embedded Software Engineer · Vecmocon Technologies May 2026 – Present · Noida

- Develop firmware on TI MCUs for Vehicle Intelligence Modules (VIM) interfacing the vehicle's BMS, motor controller and VCU over CAN, for OEM and fleet-scale EV deployments.
- Integrate Quectel EC200-series cellular modules for real-time vehicle connectivity and telemetry uplink.
- Build with reliability as a first-class requirement — rigorous testing and debugging in the development loop for field-level dependability; building working expertise in AIS-140 telematics compliance.

Embedded Software Engineer · MLworkX Pvt. Ltd. Jun 2025 – May 2026 · Noida

- Led end-to-end R&D of production embedded products for global clients — schematic design through firmware architecture to commercial deployment — across STM32, ESP32 and 8051 platforms.
- Integrated UART-based cellular connectivity (Neoway N58) for real-time IoT communication; configured peripheral drivers, UART/SPI/I2C/BLE stacks and RTOS task management.
- Led real-time debugging and validation (oscilloscope, logic analyser, JTAG/SWD) — reduced field issues by 90% and validated 2,000+ production units; optimised firmware for low power and owned board bring-up.

Embedded Software Engineer Intern · Surfytch Electronics Dec 2024 – May 2025 · Ghaziabad

- Developed C/C++ firmware on ATmega328P and STM32 with FreeRTOS; optimised interrupt handling and memory footprint on resource-constrained MCUs.
- Built a UART-based FOTA bootloader for over-the-air updates and a complete RFID access-control system (scan, verify, respond).

KEY PROJECTS

- Vehicle Telematics Device** (STM32 · Neoway N58 GSM · CAN · Jul-Sep 2025) — Live vehicle data acquired over CAN bus, telemetry routed to ThingsBoard via MQTT, FOTA over HTTP for remote firmware upgrades — the same BMS/VCU/CAN + cellular stack used in production EV telematics.
- Automatic Pill Dispenser** (Commercial product, US client · Nov 2025–Mar 2026) — STM32F103 firmware controlling 4 motors and multiple sensors for fully automated dispensing; Android S720 board integrated for UI. Delivered end-to-end, now in commercial production.
- Smart Hip Trainer** (Wearable · ESP32-S3 · Jan–Mar 2026) — Owned R&D, schematic design and firmware; Bosch BNO055 IMU for real-time hip and full-body motion tracking, streamed live to a companion mobile app over BLE.
- Hydrogen Gas Generation System** (STM32 · Jun–Aug 2025) — Relay-based motor control with RTC scheduling for autonomous generation cycles, motor–valve synchronisation and emergency safety handling.

EDUCATION

B.Tech — Electronics & Communication Engineering

United College of Engineering and Research, AKTU Lucknow
Sept 2021 – May 2025 · **CGPA 8.6 / 10.0 — Department Topper**

Certifications: Design & Fabrication of Electronic Circuits · Innovation in Circuit Engineering using IoT · Springer Conference Presentation · Letter of Appreciation

ACHIEVEMENTS

- Presented a live demo of the RedRaven™ fire detection system aboard **INS Airavat** to the Admiral and Head of Naval Dockyard, Visakhapatnam.
- Granted Indian Patent** — MIMO antenna design.
- Best Paper Award, ETMCIS 2024 (IEEE)** — Hybrid Metal-Graphene THz MIMO Antenna; 2 papers published in international journals/conferences.